

Framing Vocoder for data compression using Deep Learning



Department of Speech, Music and Hearing

Submitted By: Parul Dhingra

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Supervisor/Advisor: Professor Jonas Beskow
Professor Bernd Edler

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Summary and Conclusions

Future Scope

1. Introduction

- One of the most primal and effective medium to convey information is sound.
- Different languages have different sound libraries associated to it.
- Signalling through gestures to signalling through highly developed speech libraries. This makes the process highly efficacious.
- Speech waves are extensively incorporated in the field of science and technology and constitute the most influential part of it.
- The analysis and synthesis of voice led to the invention of Vocoders(voice coders) in 1939 at BELL Labs by DUDLEY's spectacular model of artificially generating speech.
- Compression and multiplexing techniques.

1. Introduction contd..

- Enforcing machines to learn data representations and perform tasks with **higher accuracy**, where **less manual interference** is required and **thoughtful decisions** can be made.
- In the present era, **deep learning**, which is based on **learning data representations**, is an integral part of **audio/speech recognition, computer vision, and natural language processing**.
- **Autoencoders** are modeled for **unsupervised learning**
- The thesis work aims to formulate a novel **speaker-specific vocoder**, in a competent way where **autoencoder is trained** and designed **to compress** the speech data.

1.1 Vocoder

- Sound/Speech is fundamentally **pressure waves**, made up of **peaks and troughs** which are **regions of higher and lower pressure** respectively.
- **Speech sounds** comprise of tones of several frequencies and generally can be differentiated into two categories **complex periodic sounds** and **aperiodic sounds**.
- In **human-speech articulation mechanism**, the **vibration of vocal folds** represents the **complex wave** where its **components** are called **harmonics**.

1.1 Vocoder

- The analysis of frequencies, its source of generation and filter mechanism leads to the classification of sound in broadly two major categories i.e. **voiced** and **voiceless**.
- Figure 1.1 , illustrates the waveform representation of voiced and voiceless sound where male speaker speaks letter 'X' which consists of the voiced part(eh) /E/ which is followed by the unvoiced sound /s/.

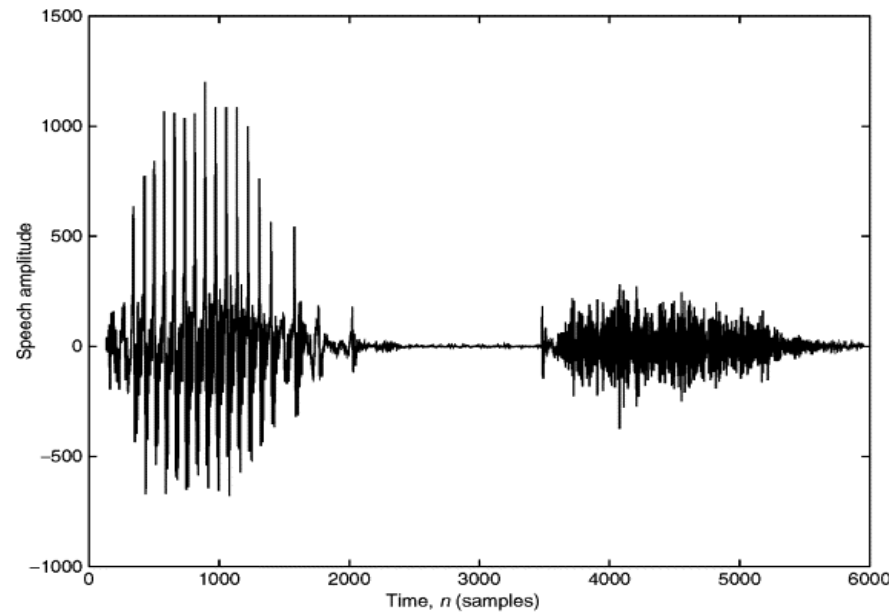


Figure: 1.1

1.1 Vocoder

Figure 1.2, speech analysis-synthesis system

- It is generated on the principle that vocal source excitation can be **voiced** or **unvoiced**.
- In the **analyzer part**, the frequency discriminator and frequency meter measured the fundamental component in **voiced sounds**. There is generally an insufficient power in the fundamental frequency range in the case of **unvoiced sounds** to operate the frequency meter.

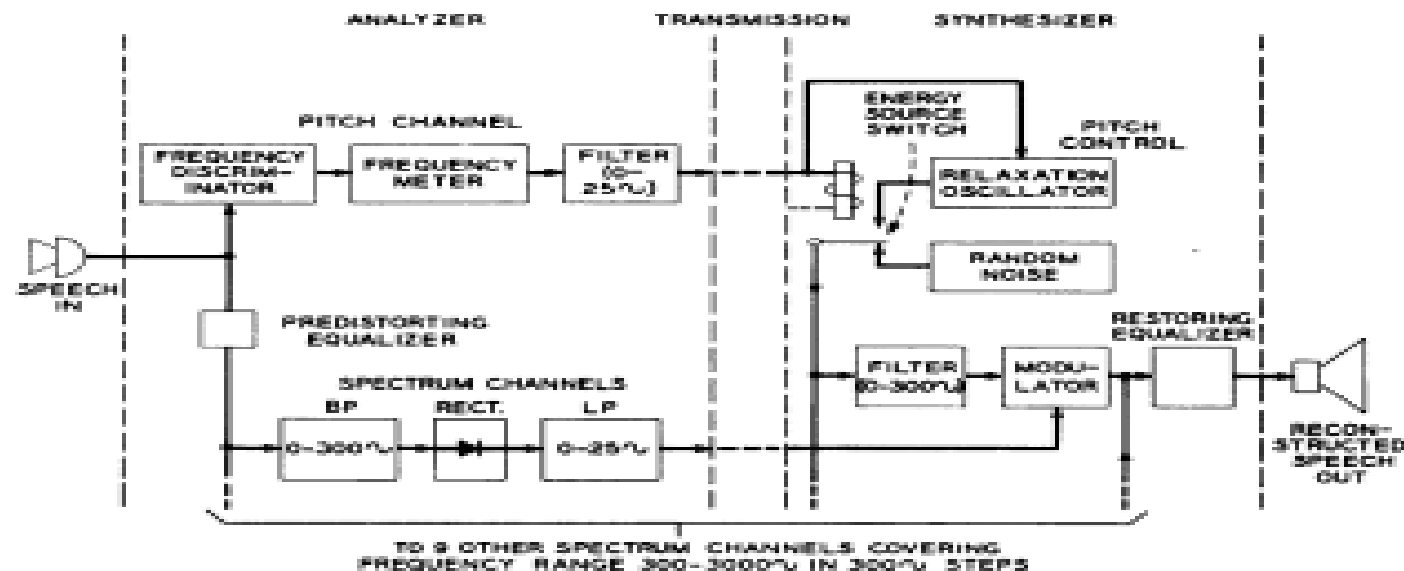


Figure: 1.2

1.1 Vocoder

- Further, the advancements of **linear prediction techniques** have led to the development of vocoders in the terms of improving the quality of speech synthesis.
- Figure 1.3, represents the block diagram of **pitch-excited linear prediction vocoder**.
- Transmitter
- Receiver
- Real time frame by frame

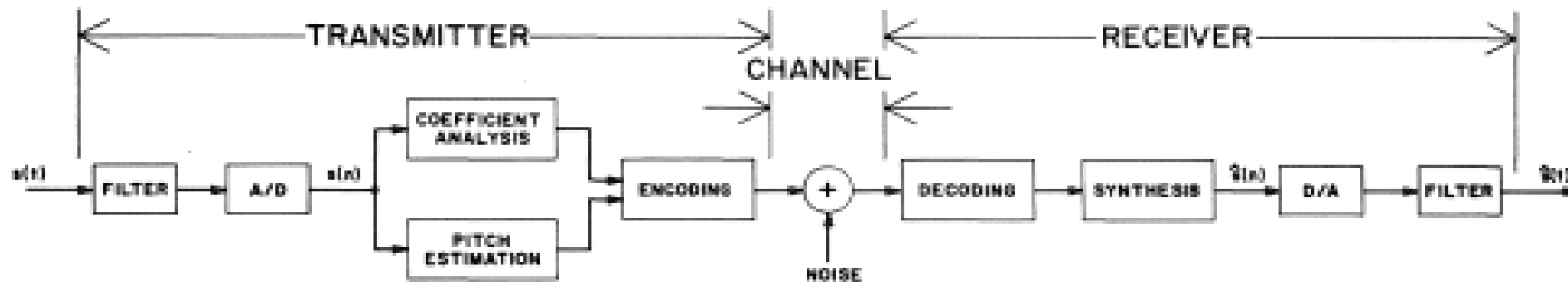


Figure: 1.3

1.2 Autoencoder

- **Neurons** form the basic blocks/elements of learning models
- The outputs are the values from the **activation function** which are fed with the weighted-sums of inputs.
- The **shallow models** can provide efficiency for simpler tasks but they are not well suited for the complex application like image/speech analysis/recognition and understanding, hence we move towards the **deep-structured models**.
- The deep learning models can be classified into **unsupervised, supervised and hybrid learning models**.

1.2 Autoencoder

- **Autoencoders models** are unsupervised learning models.

Figure 1.4, represents a simple autoencoder with one hidden layer and output/input layer.

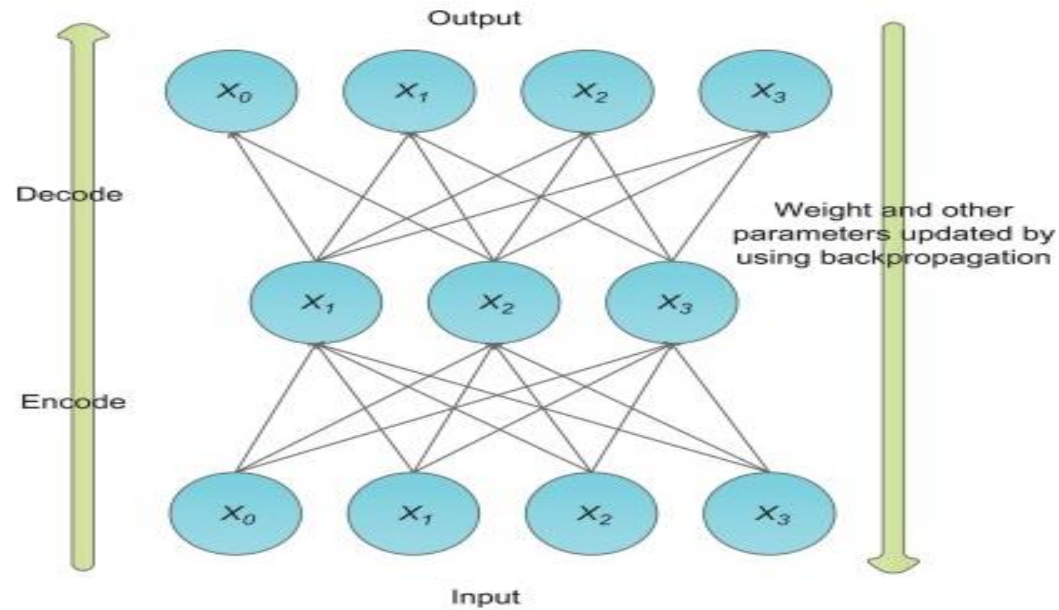


Figure: 1.4

1.2 Autoencoder

- Efficient non-linear mapping of data where its motive can be to learn an identity function.
- Encoder and decoder.
- They are trained with Backpropagation algorithm where its objective function can be to minimize the mean-square error

1.2 Autoencoder

- Backpropagation algorithm comprises **feedforward** and **backpropagation** processes.
- Figure 1.5, represents a basic framework for backpropagation algorithm.

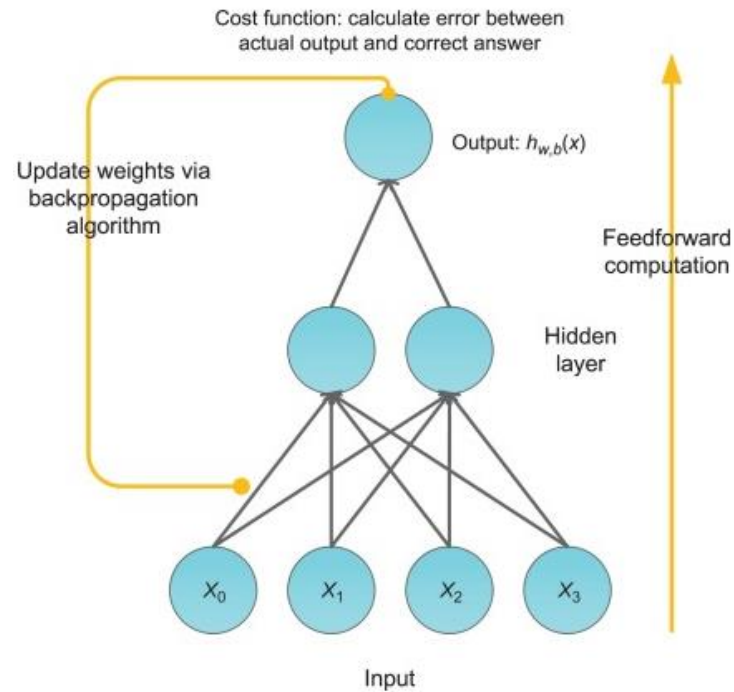


Figure: 1.5

1.3 Integration

- The exploration of **vocoders and autoencoders** presents an interesting thought of combining the **principles** of both of them **to formulate a model** that can be used **for compression of data/signal**.
- **Trained on the speech corpus** of an individual making it **speaker specific**.
- The **speech sentences/signal** are **segmented** to form **vectors of equal length** which are used to **train the autoencoder** and build a model for **data compression** for speech signals.

2. Perspectives of Speech Analysis

- There can be **numerous ways to process and analyse speech signals** e.g. articulatory analysis/processing, frequency/amplitude selective analysis/processing and many more.
- The section explains the **implementations of the specific ways** of processing the speech data which is required in constructing the model in the project.

2.1 Articulatory Perspective

- It links the **speech production mechanism** in human to the model which can generate **artificial speech**.
- Figure 2.1, shows one of the **world's first electrical speech synthesizer(Voder-Voice Demonstrator)**.

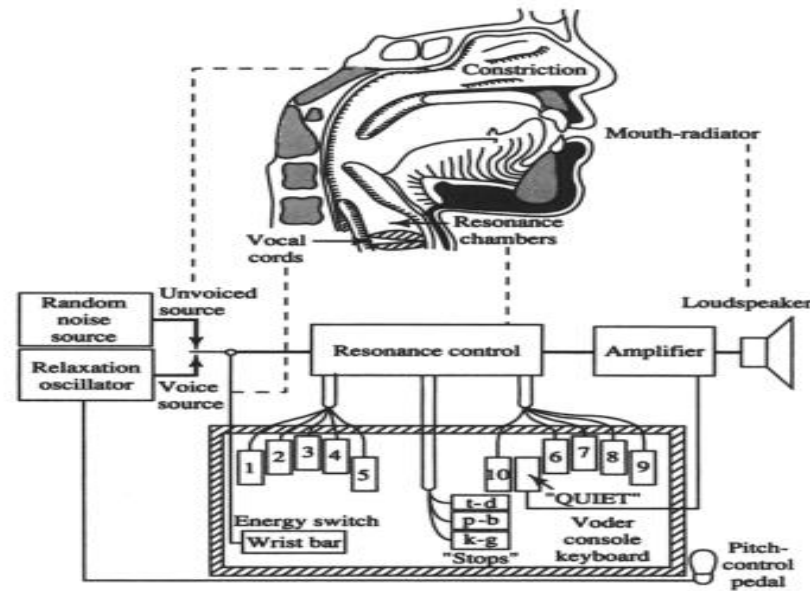


Figure: 2.1

2.1 Articulatory Perspective

- The excitation source of **voiced sound** is the vibration/**oscillation of vocal cords**.
- The source of excitation which results from the **turbulent flow of air** leads to the **unvoiced sound** in the acoustic system.
- Figure 2.2, illustrates various **formations of vocal folds** in (a) normal breathing, (b) complete closure, and (c) voicing .

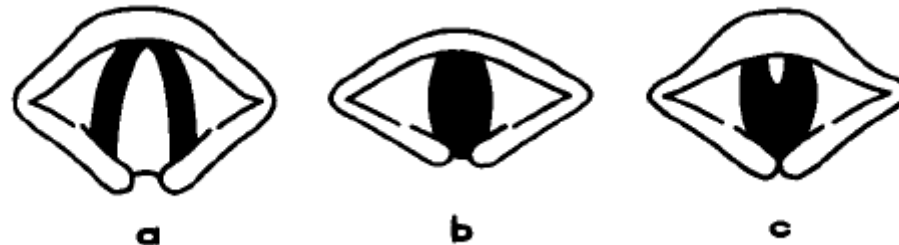


Figure: 2.2

2.2 Harmonics and Bursts Modeling

- In order to **decompose the speech signal** into **voiced and unvoiced channels**, the **harmonics and bursts modeling** is implemented.
- The **voiced and unvoiced** sounds **differ** in their **spectro-temporal properties** which is taken into account for decomposition into two channels.
- Voiced sounds **analyzed in spectrogram displays self-similarity over time**.
- For the **voiceless part** of the speech signal, it is expected **that energy is more distributed along the frequency axis**.

2.2 Harmonics and Bursts Modeling

- Figure 2.3, shows the MATLAB spectrogram implementation for the harmonics and bursts decomposition for the sentence '**Egg shell is not good to eat.**' The uppermost plot is the original speech signal, the plot in the middle represents the harmonics part and bottom most corresponds to the bursts- channel.

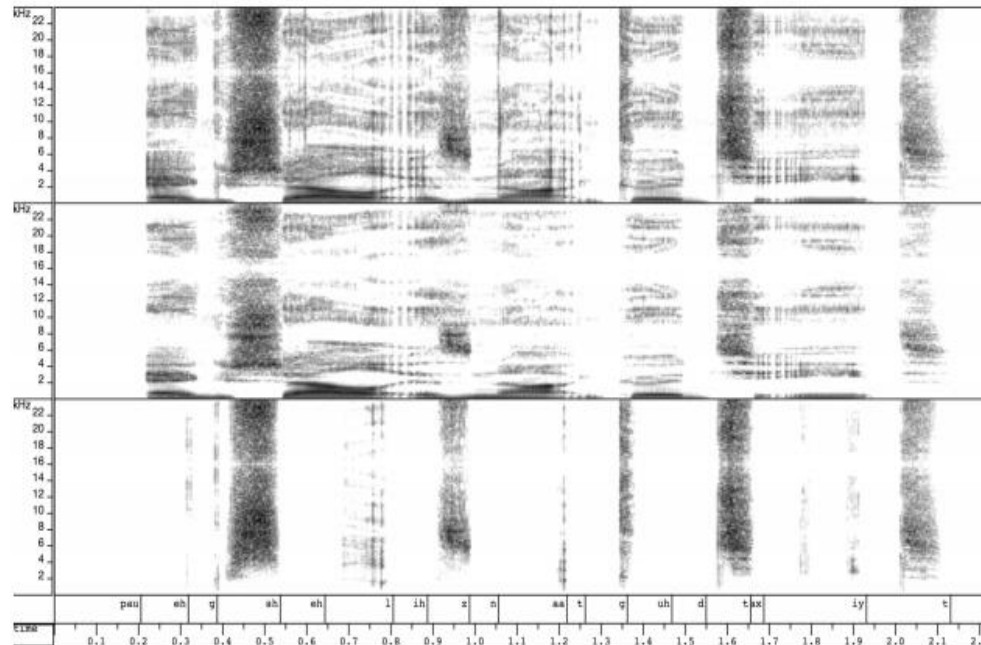


Figure: 2.3

2.3 EGG and GCI

- **Electroglottography** is a non-invasive technique to capture the glottal movements in terms of waveforms.
- In order to **capture waveform** two **metal electrodes** are kept on both side of the neck where the glottal region is located.
- The **opening and closing of vocal folds** are captured as the **measure of variation of electrical current**.
- In order to **detect the GCIs** from the EGG waveform, **SIGMA algorithm** is implemented. The algorithm is stated to give **GCI hit rate of 99.47 %**, which is excellent to build the new project in which GCI detection is required.

2.3 EGG and GCI

- Figure 2.4 and Figure 2.5, shows the speech waveform, EGG and **GCI**s marked with 'x' for sentences 'Lord, but I'm glad to see you again, Phil.' and 'It's the aurora borealis.' when they are analyzed in MATLAB respectively. The stereo signal containing speech waveform and EGG is provided by FESTVOX-CMU Arctic Database which is recorded at 16bit 32KHz, in a soundproof room. The implementations are performed in MATLAB environment.

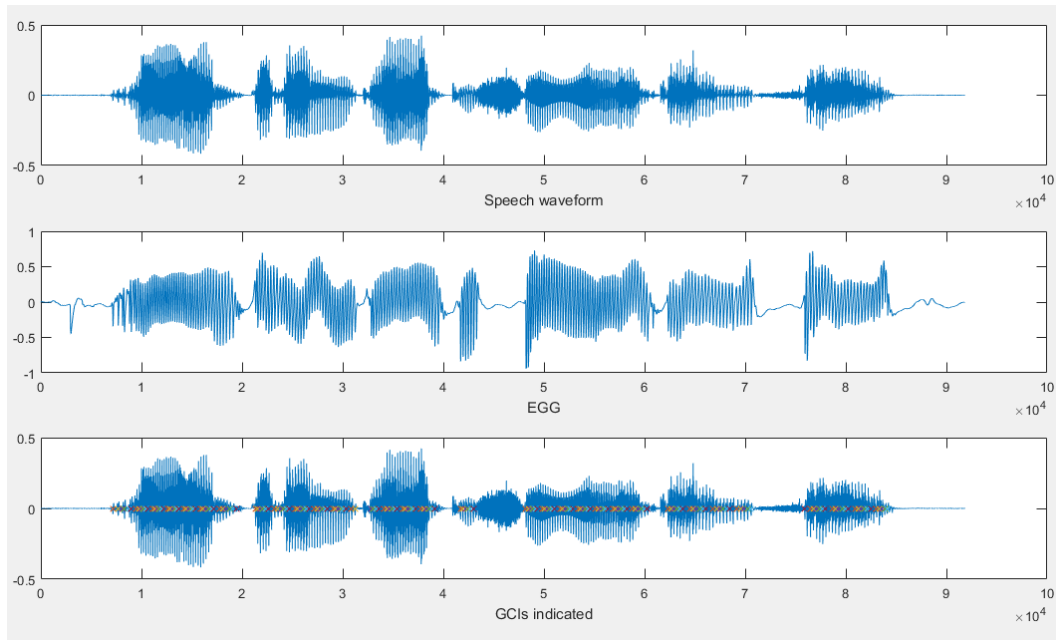


Figure: 2.4

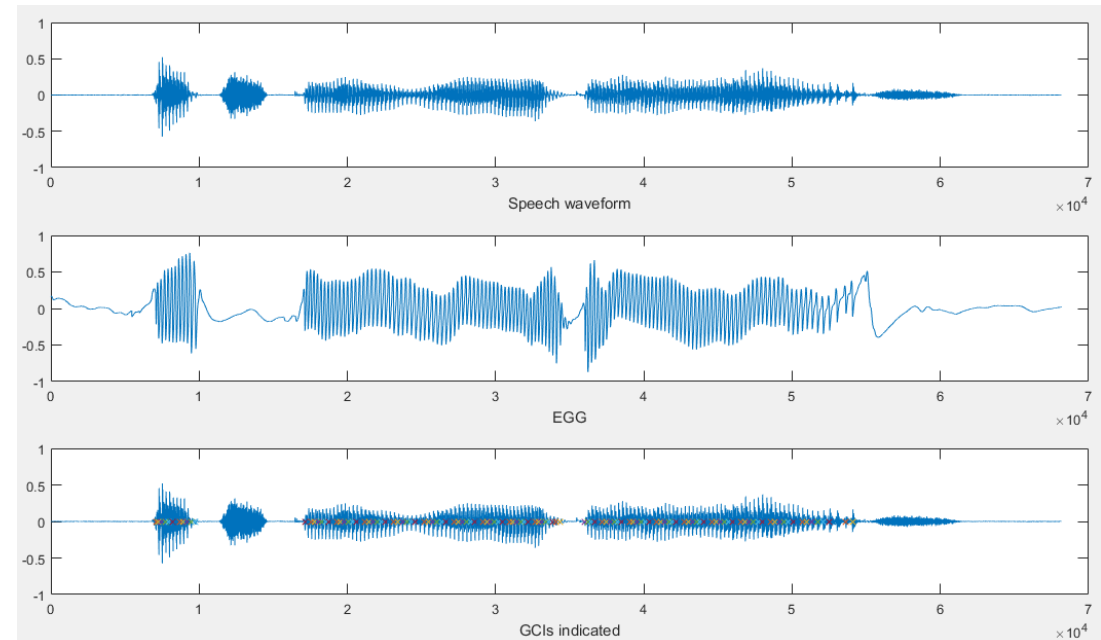


Figure: 2.5

2.4 Integration

- The **human voice physiology** gives the explanation about the **voiced source excitation where the vocal folds vibrate**. The vibration represents the complex periodic waves. These waves are further filtered through vocal tract which is frequency selective. In the case of unvoiced sounds, the source of excitation is rather a noise source.
- In order to **decompose the speech signal** into voiced and unvoiced sound channel, **harmonics and bursts modeling** is performed.
- For **detecting GCIs in the voiced channel**, **SIGMA algorithm** is implemented on EGG signal. The information about GCIs is used to construct the segments required to train the autoencoder.

3. HBEWseg- framework for speech segmentation

- HBEWseg-Harmonics, Bursts and EGG Waveform based segmentation and reconstruction of the speech signal is proposed, described and analyzed.
- The database on which implementations are performed is provided by **FESTVOX-CMU Arctic Library**.
- There are **1132 utterances by US English male speaker**, which are recorded in the soundproof room, at **16bit-32KHz**. For each sentence, a stereo signal is present, in which **first channel/column** is **speech waveform** and the **second channel/column** is **EGG signal**.

3.1 Framework Overview

- The framework has **two main components of framework** namely **preprocessor** and **reconstructor**.
- For the **voiced** channel/part, these **instances** are used to **create segments of equal length** at the preprocessor.
- The **reconstructor**, receives these segments, the length of the original speech signal and a vector containing glottal closure instances to **regenerate the voiced part of the signal**.
- The **bursts channel/vector is overlaid** on the harmonics channel and the speech signal is generated back from segments.

3.1 Framework Overview

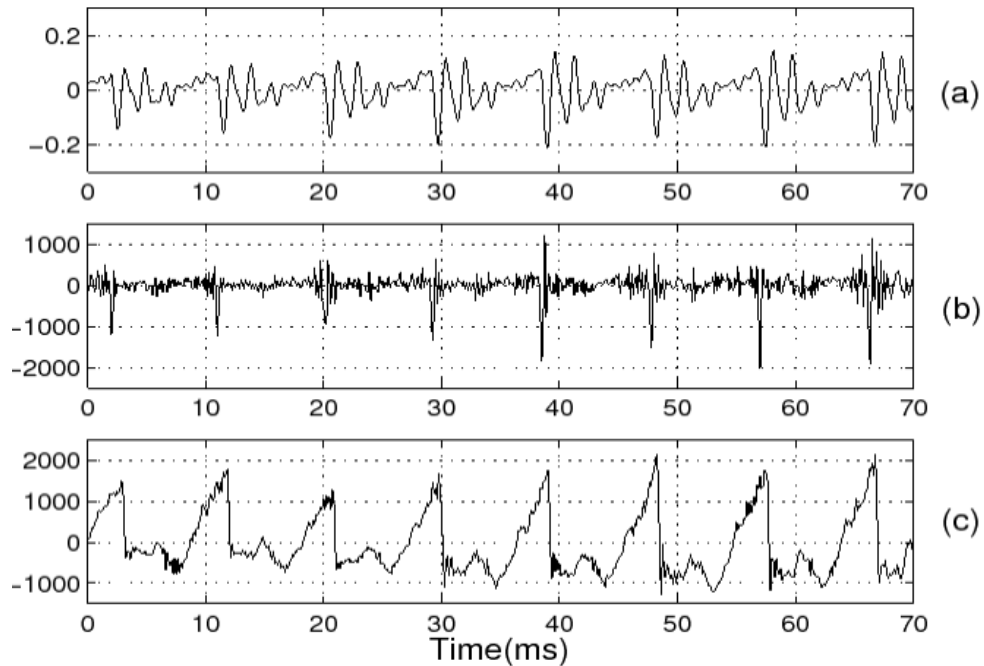


Figure 8: (a) Segment of voiced speech /a/, its (b) LP residual and (c) glottal pulse waveform.

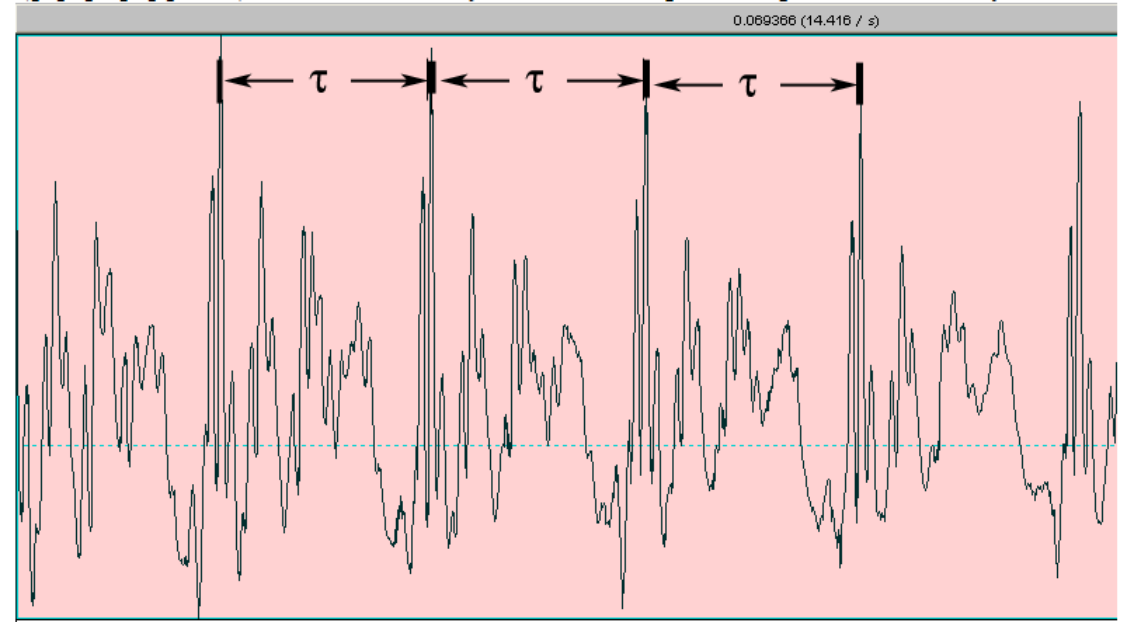


Figure: 3.1 : Analysis for segmentation(not plotted by me; it is for reference to understand the concept)

3.2 Description and Implementation of Preprocessor

The framework is analyzed with two utterances in the sections below in order to properly **state the implications**. The first utterance(**Case A**) is **"They were following the shore of a lake."** and the second utterance(**Case B**) is **"It was not a large lake, and almost round."**

Step 1

Figure 3.1, Figure 3.2, shows the speech waveform and EGG signal on which implementations are described for case A and case B respectively.

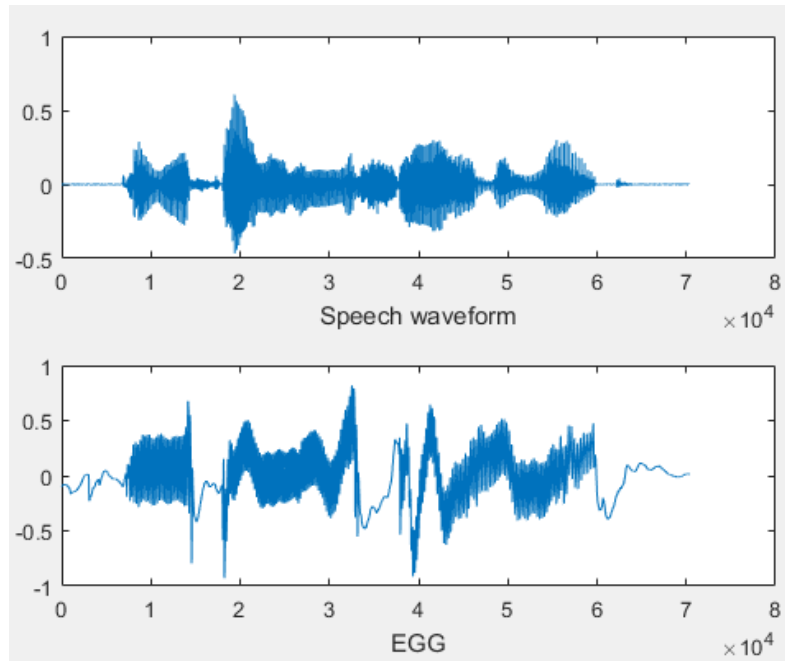


Figure: 3.1

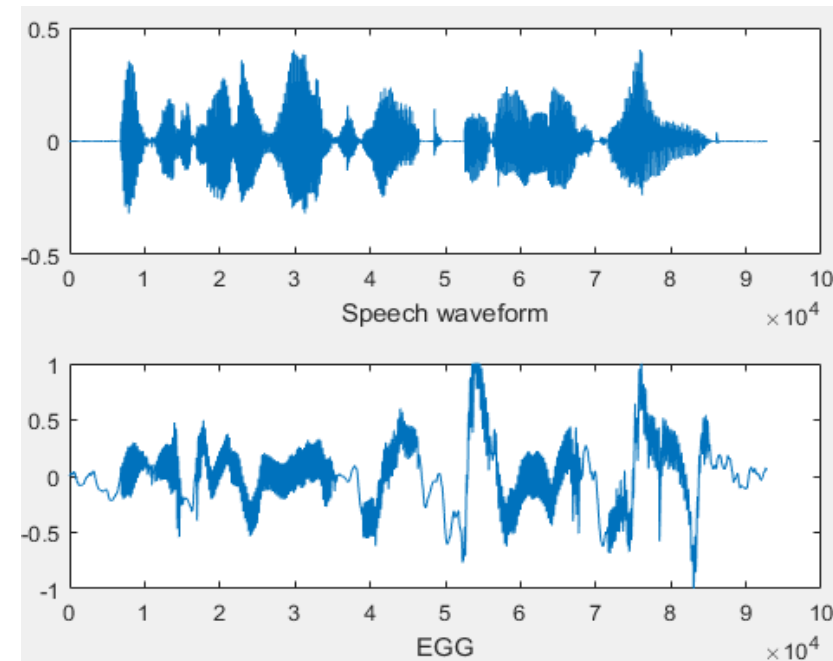


Figure: 3.2

3.2 Description and Implementation of Preprocessor

Step 2

The EGG signal is analyzed and GCIs are detected in the waveform. An **information matrix** is created which contains the information about GCIs. The **first column** contains the GCIs and **second and third column** contains the **difference of it with respect to previous and next GCI**. This matrix is used for analyzing the segment generated. Figure 3.3, Figure 3.4, shows the GCIs marked for both the cases for voiced sounds with 'x'. Further, the speech signal present in the first channel is decomposed into harmonics and bursts channels.

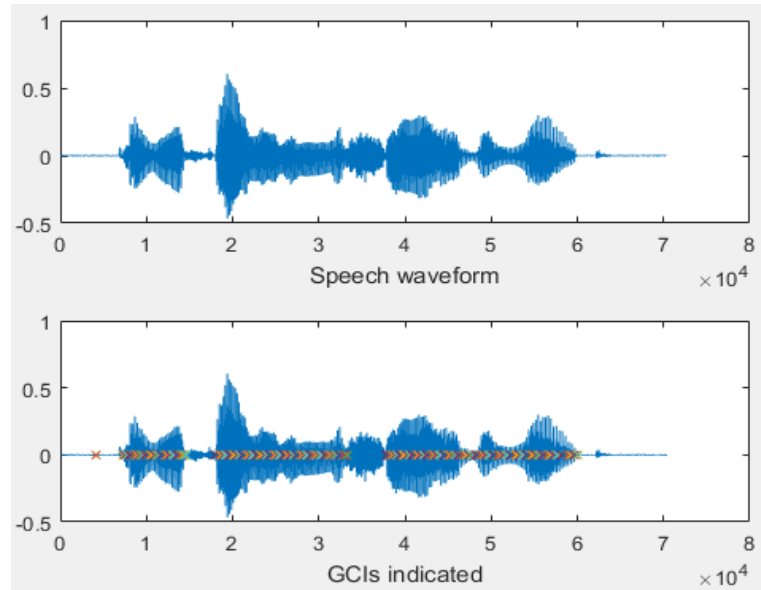


Figure: 3.3

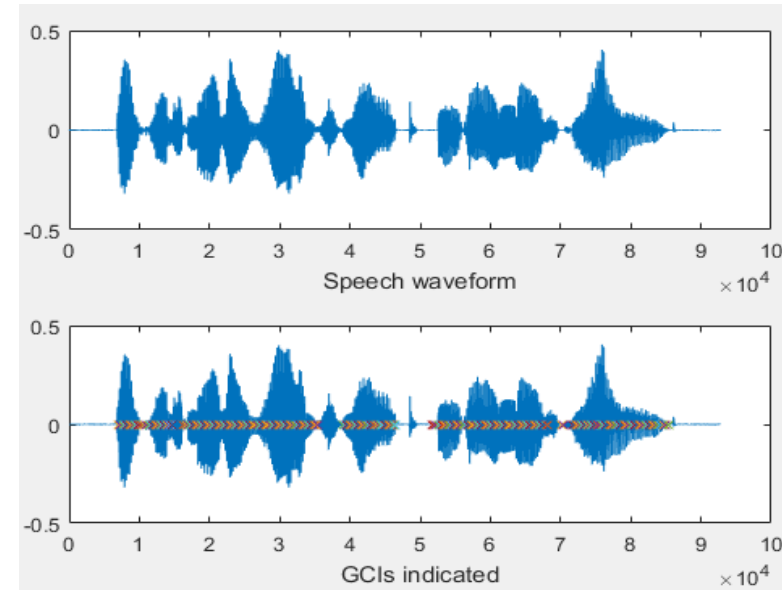


Figure: 3.4

3.2 Description and Implementation of Preprocessor

Step 3

- The **studying of the information matrix** gives the idea about what the length of the segments can be.
- The framework is currently designed for **male speakers** where the **time duration is comparatively large** between one GCI to another as compared to female speakers.
- In order to generate a segment, **signal containing two-pitch periods is extracted, Hann-windowed with variable length windowing function, interpolated with zeros to make them of equal length of 1061 samples** which are stored for further processing. The segment describes the variation of the pitch pulses that includes both the voice source parameters and the vocal tract/filter parameters(formants).
- For **every segment**, the sample in the **middle, 531st sample**, corresponds to value corresponding to the **current GCI**.
- The cases corresponding to **consecutive voiced and unvoiced or unvoiced and voiced** are discarded.

3.2 Description and Implementation of Preprocessor

- Figure 3.5, Figure 3.6, shows the block of first 100 segments generated for case A and case B respectively.

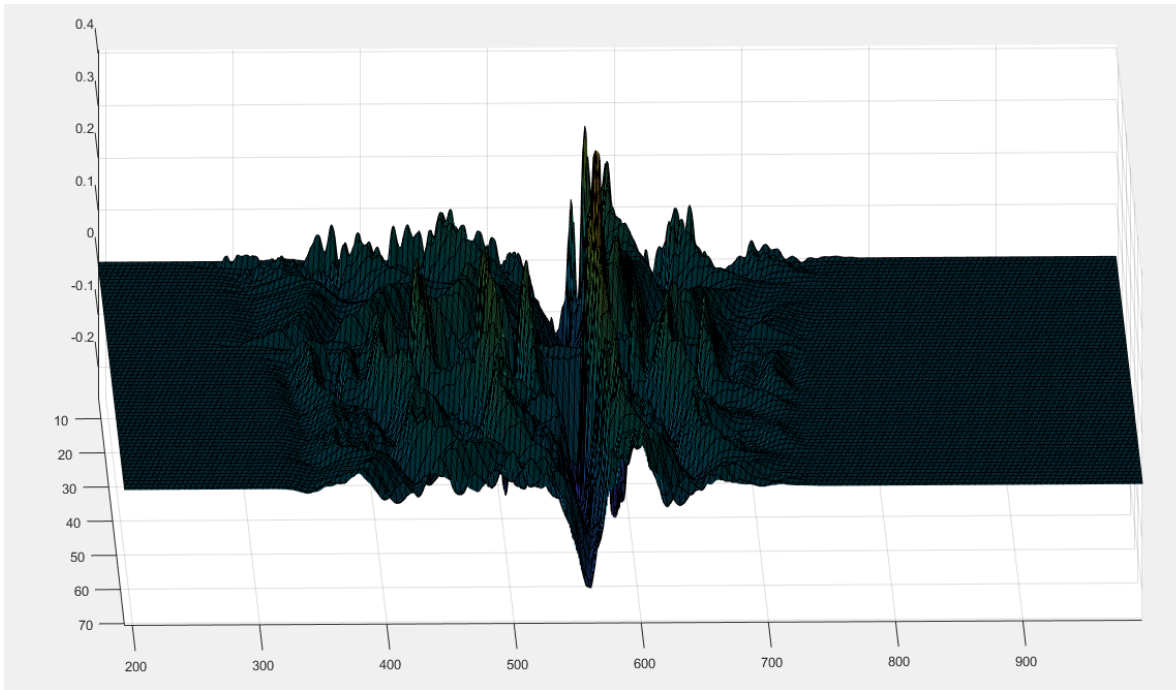


Figure: 3.5

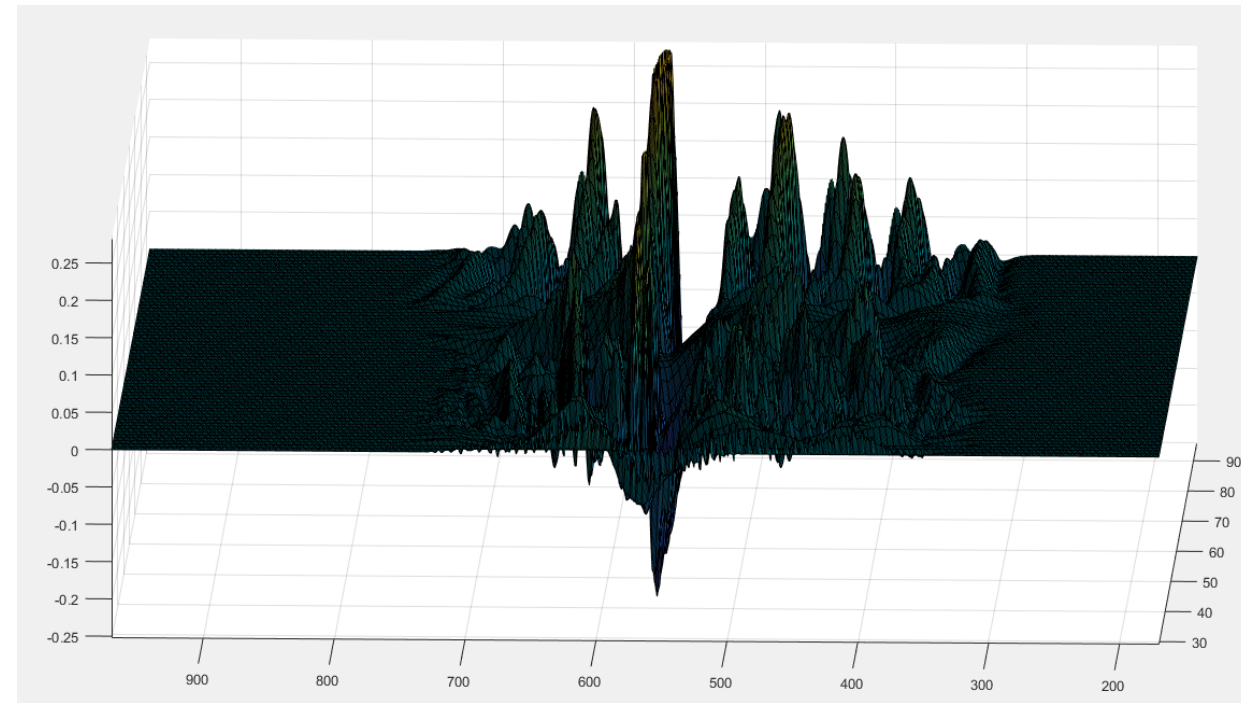


Figure: 3.6

3.2 Description and Implementation of Preprocessor

Step 4

- The **interpolated segments with zero** when displayed in a vector **consecutively** one after another are shown in Figure 3.7 for case A and case B respectively.
- **GCI**s and **segments** stored in the **same order** as they occur when an utterance of a sentence is made.

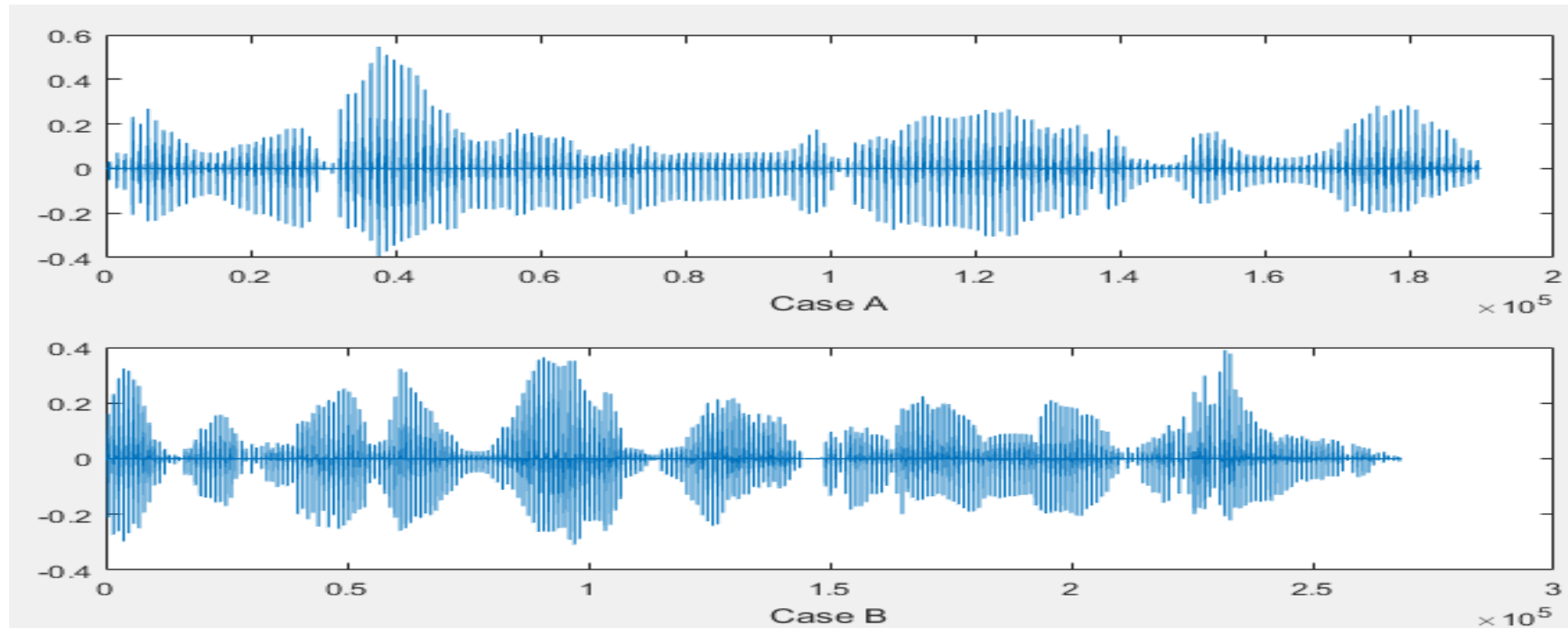


Figure: 3.7

3.3 Description and Implementation of Reconstructor

- The **reconstructor regenerates** the speech signal **from the block of segments**.
- A **zero-vector** is initialized with **length equal to** the number of **samples of the speech signal**.
The **segments corresponding to each GCI is overlapped and added** in such a way that the subjective quality does not degrade.
- **Bursts** channel is **directly added** to the vector of reconstructed harmonics channel.

3.4 Quality Analysis

- Highest possible quality thresholds before fitting autoencoder model.
- The analysis shows that the reconstruction is very promising and the framework can be used to further fit the autoencoder model to it.

Figure 3.8, shows the original signal(uppermost) and reconstructed speech signal(bottommost) for case A.

Figure 3.9, shows the original signal(uppermost) and reconstructed speech signal(bottommost) for case B.

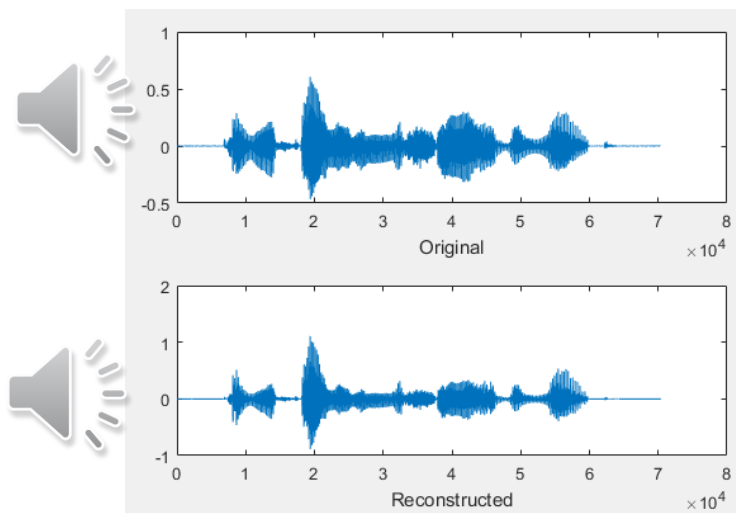


Figure: 3.8

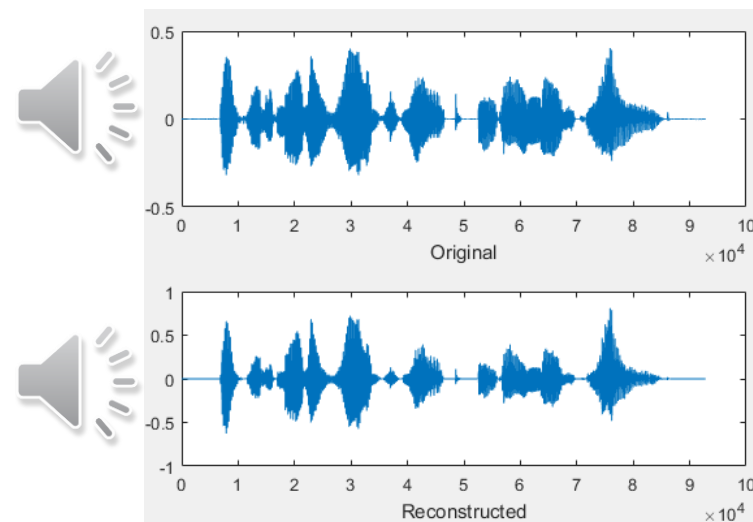


Figure: 3.9

3.4 Quality Analysis

Figure 3.10, shows the spectrogram of original signal(uppermost) and reconstructed speech signal(bottommost) for case A.

Figure 3.11, shows the spectrogram of original signal(uppermost) and reconstructed speech signal(bottommost) for case B.

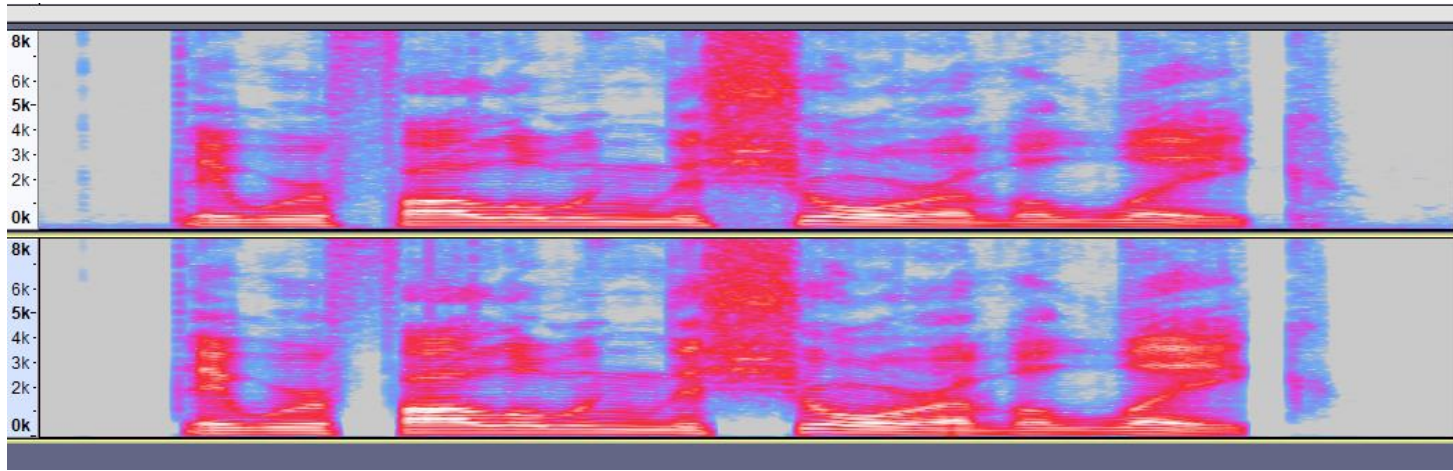


Figure: 3.10

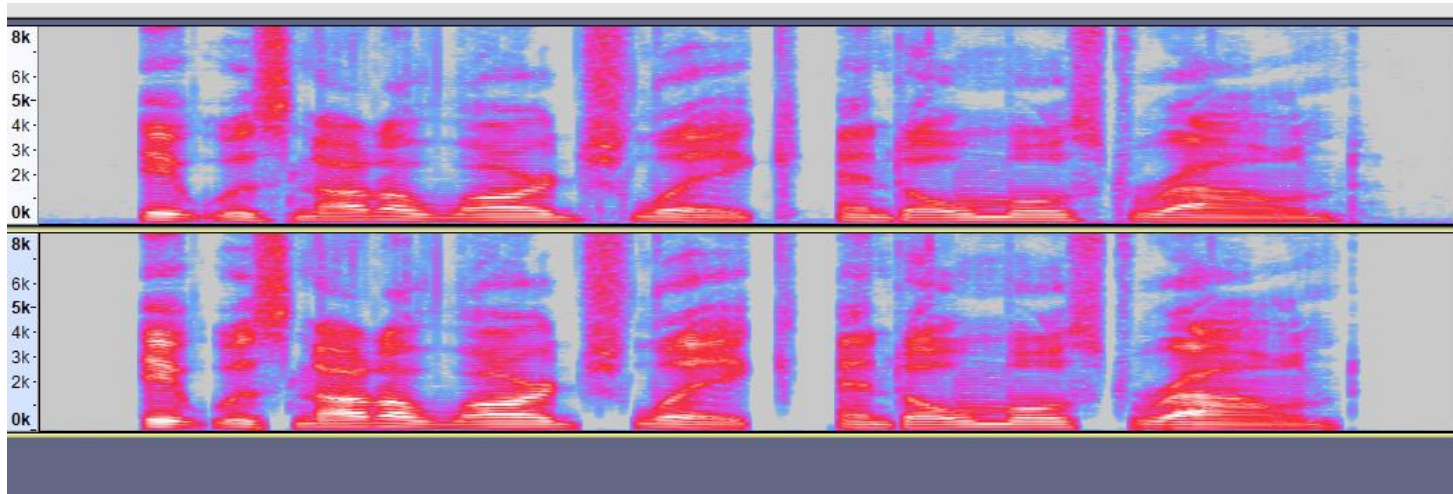


Figure: 3.11

3.4 Quality Analysis

Subjective quality assessment

Experimental Setup:

Key points:

Listening Test

Crowdfunder platform

Source of Reference files: Festvox: CMU_ARCTIC Database(male- English speech files) at 32kHz – mono channel

Countries launched in with expected high hit rate of knowing English per listener: United States of America , Sweden, Norway, Australia, United Kingdom, Germany, Netherlands, New Zealand, Canada, Denmark

Listeners experience level: Level 2- medium (experience level on the scale of 1 to 3)

Total- 500 judgments

Category- Reference audio file, Processed audio file

Objective:

The objective of listening test was to measure the degree of degradation in processed files with respect to the reference files on the scale of 1 to 5 ranging from degradation is inaudible(5), Degradation is audible but not annoying, Degradation is slightly annoying, Degradation is annoying to degradation is very annoying(1).

Anchors:

Degradation by AAC- files subjected to 10kbps transmission rate—2 processed files for 2 test cases

Reference file- 2 processed files same as reference files for 2 test cases

3.4 Quality Analysis

Results:

The resulting analysis showed that 95-100% judgments were in the favour of rating it to the level 5 and 4. This gives a promising result to further work on this segmentation and reconstruction framework and to further train the autoencoder with these segments. Figure 3.12, shows the graphical representation of judgements.

Total- 500 judgments, eliminated judgments= $80+40=120$, remaining judgements=380

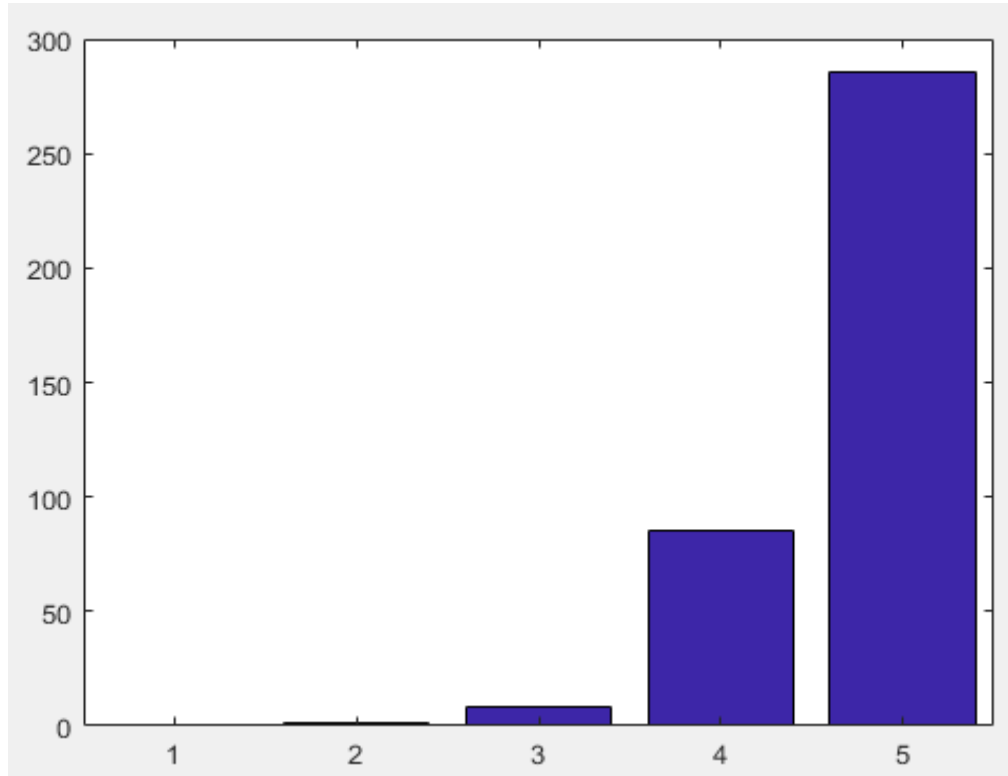


Figure: 3.12

Total valid judgements: 380

Level 5 : 286

Level 4 : 85

Level 3 : 8

Level 2 : 1

Degradation is inaudible(5), Degradation is audible but not annoying(4), Degradation is slightly annoying(3), Degradation is annoying(2), Degradation is very annoying(1)

4.1 Building Autoencoder Model

- The **convolutional autoencoder network** is implemented.
- The **compression** that takes place with autoencoders is **data specific and lossy**.
- The **compression ratio** for each utterance **depends on the number of GCIs** present in it.
- The model is **configured** to use **mean squared error loss**.
- **TensorBoard plotting**
- It is observed that **while training** autoencoder with the segments, **train loss is always lesser than test loss for every epoch**.

4.1 Building Autoencoder Model

Figure 4.1, shows the model of autoencoder that has been implemented for training and testing the segments.

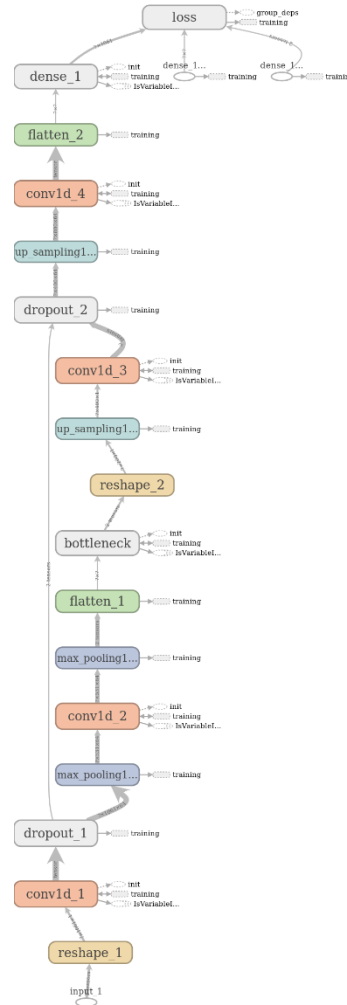


Figure: 4.1

4.1 Building Autoencoder Model

Figure 4.2, shows the training loss for encoding dimension 100.

Figure 4.3, shows the test loss for encoding dimension 100.

loss

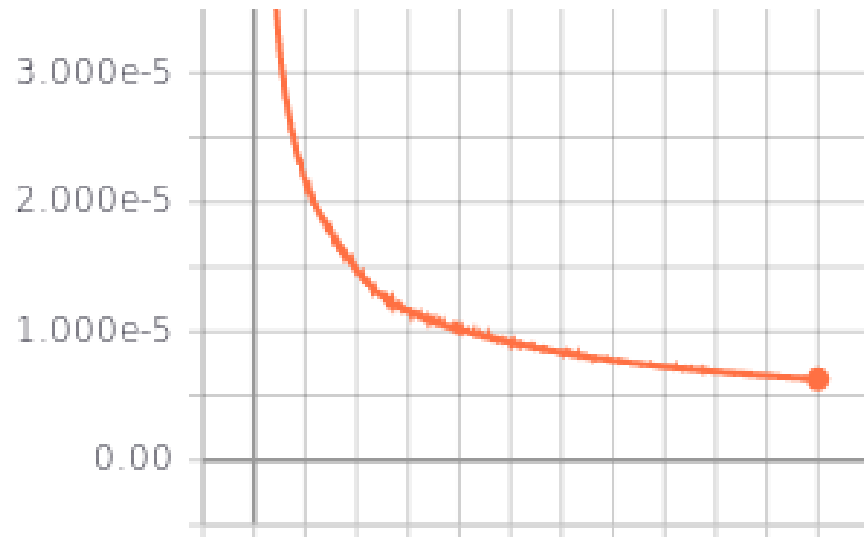


Figure: 4.2

val_loss

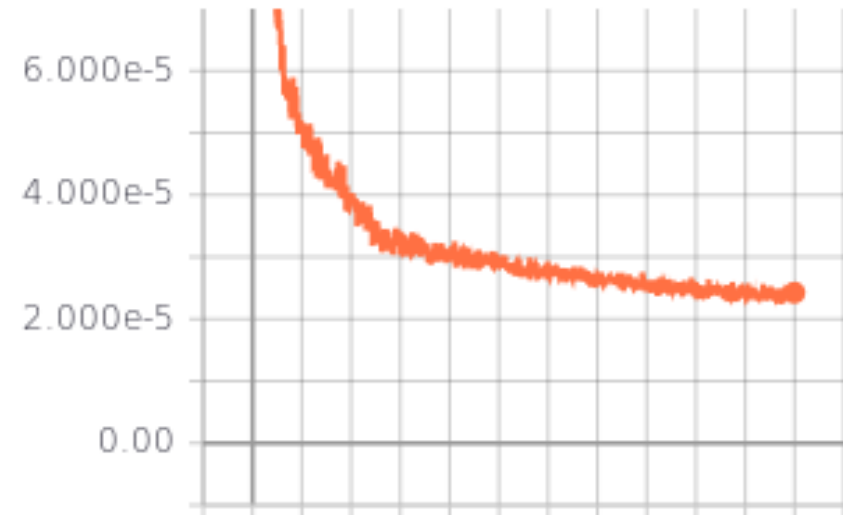


Figure: 4.3

4.2 Analysis with different encoding dimensions

- **Tested till 100 epochs** initially and the reconstructed speech files were checked at it for different encoding dimensions.
- **Considerably good quality in $10e-5$ range.** Test loss lower than that is reasonably good for reconstruction. So, this gives the analysis of the range of losses.

Figure 4.4, shows the training loss for encoding dimensions 50,40,30,25,20,16,12,8,5,4,3,2.

Figure 4.5, shows the testing loss for encoding dimensions 50,40,30,25,20,16,12,8,5,4,3,2.

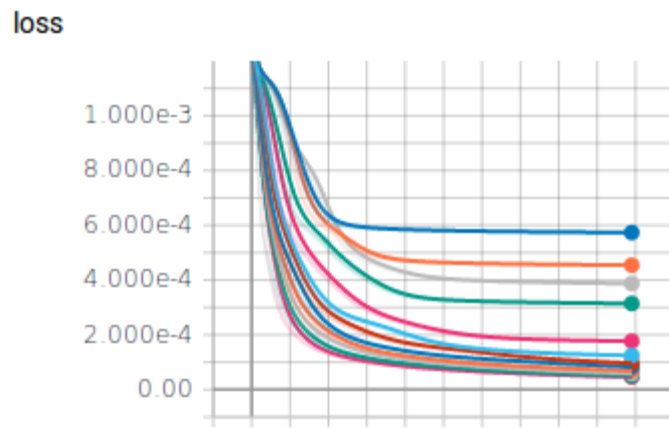


Figure: 4.4

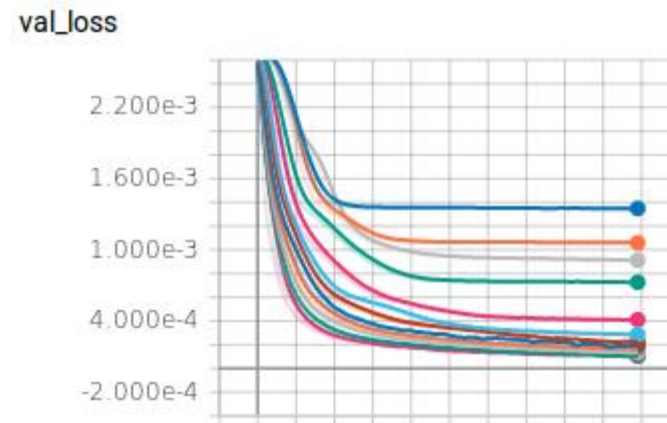


Figure: 4.5

4.2 Analysis with different encoding dimensions

- Increasing the number of epochs to 1500.

Figure 4.6, shows the training loss for encoding dimensions 50,40,30,25,20,16,12,8,5,4,3,2 at 100 and for **12 and 25** on further training.

Figure 4.7, shows the testing loss for encoding dimensions 50,40,30,25,20,16,12,8,5,4,3,2 at 100 and for 12 and 25 on further training.

The losses could be considerably brought down in the order of **10e-5** for **encoding dimension 25**. For the dimensions lower than that, the reconstruction of the signal was not proper and sound quality degraded heavily.

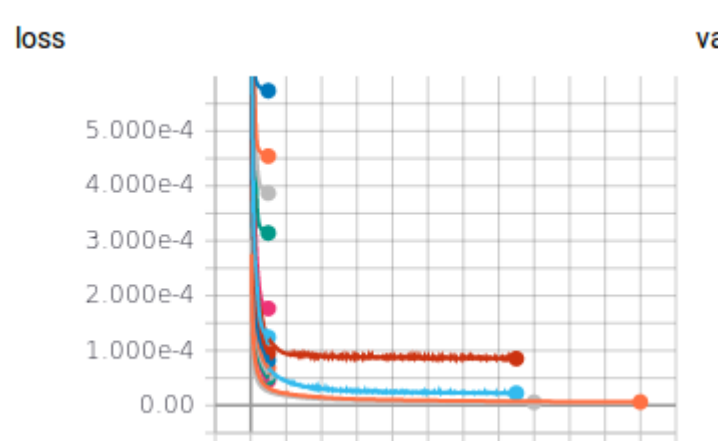


Figure: 4.6

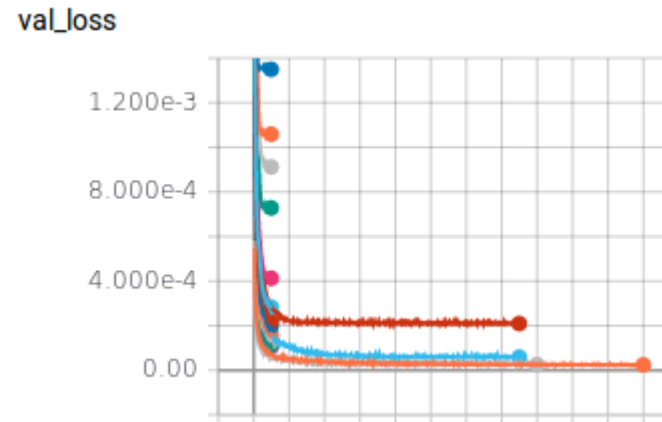


Figure: 4.7

4.3 Conclusions

- For analysis, the waveforms and their spectrogram at different encoding dimensions are plotted. It can be seen that they are reconstructed with reasonably good quality. The utterance **"He cried, and swung the club wildly."** is analysed (length of 85761 samples, number of GCIs= 242, number of GCIs after discarding specified cases= 226).
- **Figure 4.8**, shows the waveform for harmonics without sending it through the autoencoder model (uppermost), with encoding dimension **100 and 25 (bottommost)**.
- **Figure 4.9**, shows the spectrogram analysis for harmonics without sending it through the autoencoder model (uppermost), with encoding dimension 100 and 25 (bottommost). The compression for harmonics channel for each utterance/sentence is based on the number of GCIs locations present in it.
- Equation 4.1, shows the **mathematical formulation** to calculate the **compression ratio** of a particular utterance. On analyzing the cases/sentences, it is observed that the compression can be done in such a way that only 5 to 20 % of information/samples are required to be sent to reconstructor to reconstruct the signal with considerably good listening quality.

$$\frac{\text{Total number of samples in an utterance/sentence}}{\text{Number of GCIs} \times \text{encoding dimension} + \text{Number of GCIs} + 1}$$

Annotation:
Compression ratio = Inverse of
the formula expressed in Equation 4.1

Equation: 4.1

4.3 Conclusions

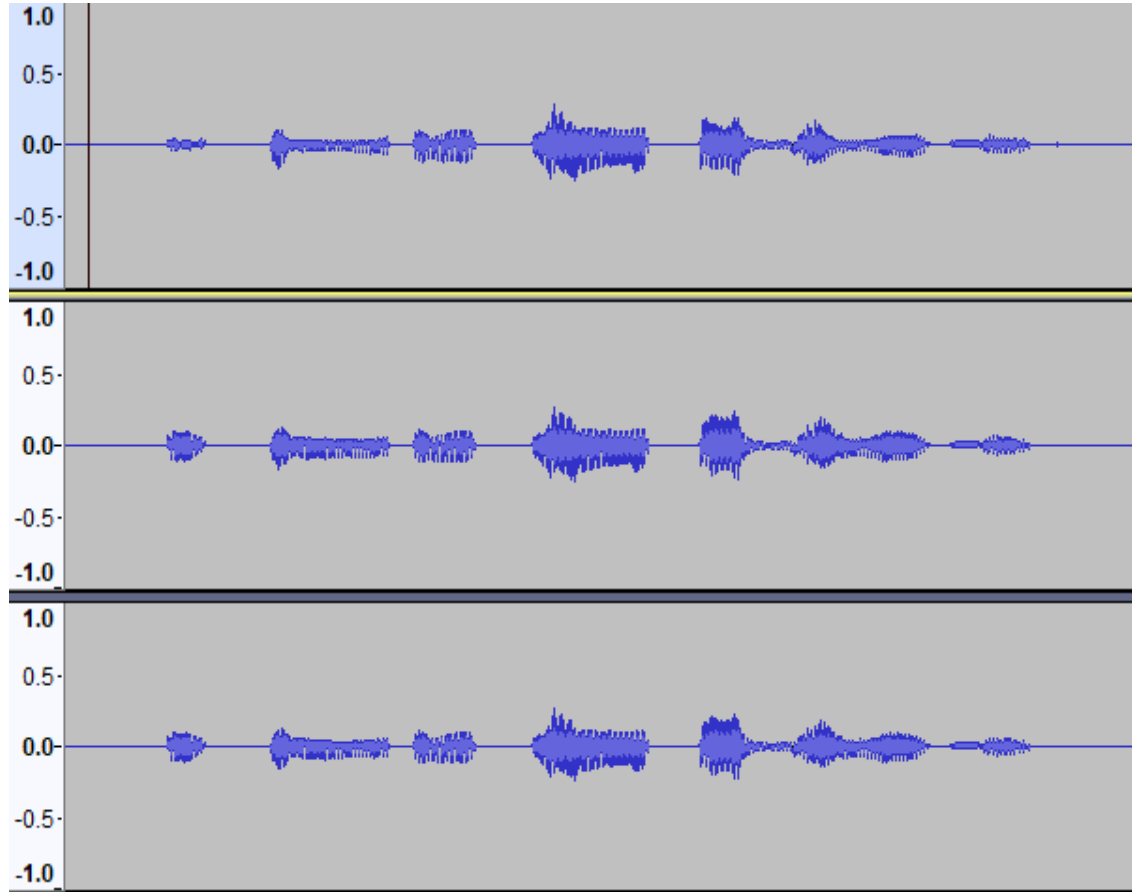


Figure: 4.8

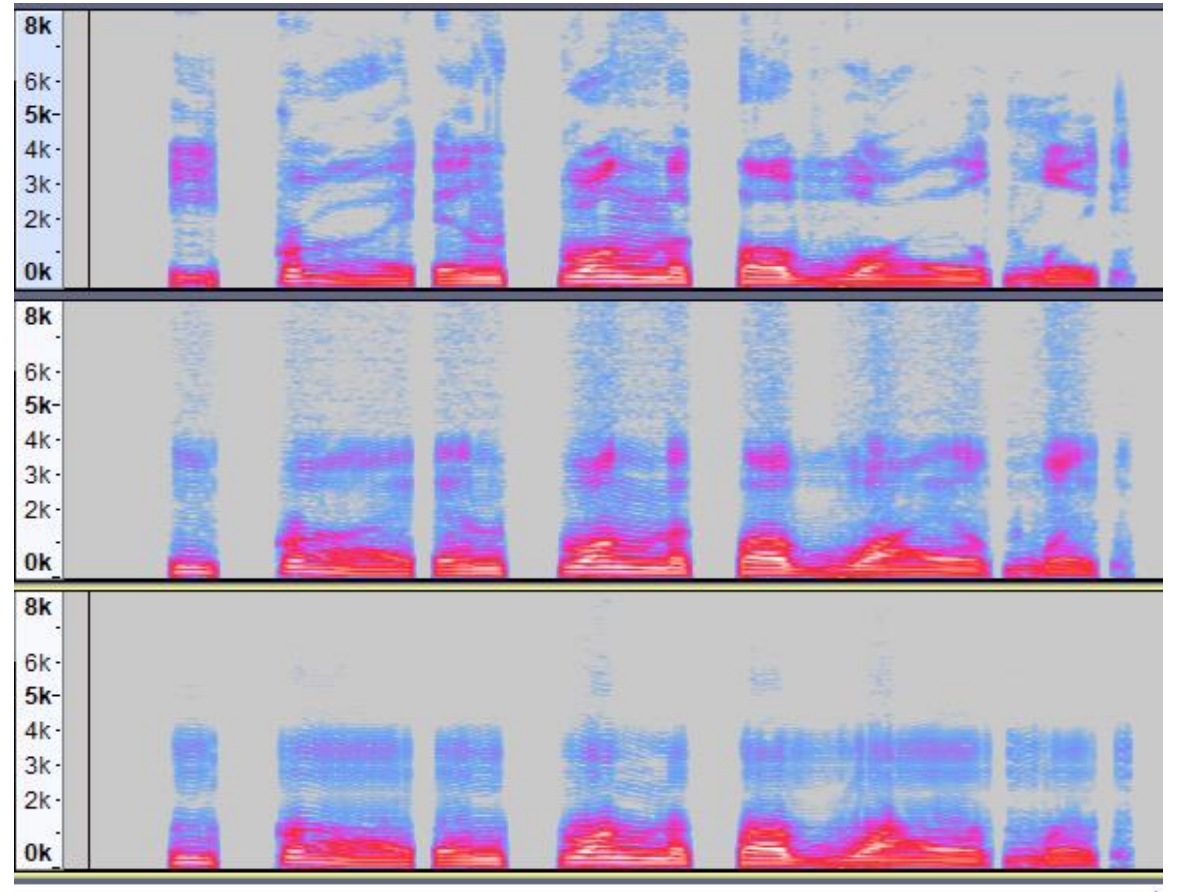


Figure: 4.9

4.3 Conclusions



Audio 1 : Reconstructed Harmonics-
without autoencoder



Audio 2 : Harmonics-with autoencoder-encdim100



Audio 3 : Harmonics-with autoencoder-encdim50



Audio 4 : Harmonics-with autoencoder-encdim25

Summary and Conclusions

- The **voiced and unvoiced sounds** in speech are **disintegrated** and treated separately.
- The framework is **speaker-specific**.
- The database consisting of **1132 utterances** by **US male English speaker**, recorded at **16bits-32KHz** provided by **FESTVOX-CMU Arctic library** is used for designing the framework.
- At the preprocessor**, the stereo signal which contains **speech waveform** in one channel and **EGG** in another is analyzed. The **EGG signal** is used to **detect the GCIs** present in the utterances. With the information about these GCIs, **segments of equal samples/length for the voiced sounds** are generated.
- These **decoded segments** are **reconstructed back** into the speech signal by **reconstructor** with the **overlapping and adding technique**.

Summary and Conclusions contd..

- The framework for speech segmentation and reconstruction **without incorporating autoencoder model** is tested **subjectively on CrowdFlower platform** and the results makes the **framework promising** to further **fit the autoencoder** model to it.
- The **segments are fed** as an input to the encoder side of **autoencoder**, which **reduces the dimension of it at bottleneck layer**, hence leading to **compression of data**. These compressed form of data/segments can be **stored or transmitted** according to the requirement.
- The **decoder side of autoencoder decompresses** these segments **back to the original length** of segments.
- The **voiced part is modeled with autoencoder** and at the reconstructor, the **unvoiced part is added** on it to generate the speech signal wholly.
- The **compression** for each utterance is **based on the number of GCIs instances**.

Summary and Conclusions contd..

- For different encoding dimensions, it is observed that the **decoded segments from autoencoder degrade in quality** when the **encoding dimension is gradually reduced**, as the **reconstruction loss is higher**.
- This analysis gives information about **limitations to how much compression** can be done without severely degrading the subjective quality of speech signal.

Future Scope

- The proposed thesis work is focused to provide **a new dimension to the field of speech technology**, enforcing a huge scope of further exploration in this direction.
- Design the bursts containing channel with the autoencoder** by studying the properties of bursts extensively.
- In future, **further formal listening tests** have to be performed **after** modeling the autoencoder with **bursts**.
- To **study the lower dimensional representations** of segments that autoencoder has learned during the training process.
- Run in real time for **speech-transmission with as low delay as possible**.
- Variants of autoencoders to further update and improve the model in future.

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Thank You.